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An edge-based method for effective abandoned luggage detection in complex surveillance videos

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ABSTRACT

Abandoned objects detection is one of the most challenging tasks in intelligent video surveillance systems. In this paper we present a new method for detecting abandoned objects (AO) using edges instead of pixel intensities. Our main focus, is on reducing false alarms, while keeping a high positives detection rate. Based on edge information, the proposed method reduces errors rate considerably compared to pixel intensities based approaches. At first, static edges are detected by applying a temporal accumulation step using the foreground edges mask resulting from the edge-based background subtraction model. Then, edges clustering is applied on the obtained stable edges mask, using edges' position and stability in time to delineate the object bounding box. Finally, an efficient classification approach, relying on edges' position, orientation, and staticness based scores, is applied on the AO candidates. The proposed approach has been validated on several challenging benchmarks, and is compared to other works of the literature. The results shows that our method reduce false alarms rates greatly, while keeping a good detection accuracy of true positives.

1. Introduction

Due to security concerns in the last decades, implementing an intelligent video surveillance system to help the human actor in monitoring public places has become a necessity. Abandoned object detection is one of the most important and critical tasks in such systems. In transit zones (airports, train stations), a luggage left by its owner is considered as a potential threat. Using a classical video surveillance system makes the task very challenging for the security staff. Firstly, it is impossible for a human operator to keep a permanent watch on the screens. Secondly, with many screens to monitor, even if the operator is actively watching, such events may not be spotted on time, and therefore, it will be too late to take action. Thus, deploying an automatic system may help monitoring public places by reducing the security staff effort and detecting abandoned objects in real-time, which gives the ability to react on time. An abandoned object, also referred to as static or as left object, is defined as a foreground object, usually a luggage, which remains static for a defined period of time.

Static object detection is the main challenging task in this research field, where it is hard to achieve a high Recall while keeping the false alarms rate low. Recall is very critical since the system will be used for security purposes. However, Precision is less critical but it is very important to keep it high, since too much false alarms would result in a heavy handicap. An effective system must therefore, overcome the issues of occlusions in a crowded environment, illumination changes and ghosts caused by removed scene objects.

Many works devoted to this task were proposed (Auvinet et al., 2006; Evangelio and Sikora, 2010; Guler et al., 2007; Hassan et al., 2013; Kim et al., 2014; Krahnstoever et al., 2006; Liao et al., 2008; Lin et al., 2015; Ortego and SanMiguel, 2014; Pan et al., 2011; Porikli et al., 2007; Martínez-del Rincón et al., 2006; Szwoch, 2016; Tian et al., 2011; Wahyono et al., 2016; Wang et al., 2014). Most of them are based on background subtraction. Some of these methods (Auvinet et al., 2006; Guler et al., 2007; Krahnstoever et al., 2006; Martínez-del Rincón et al., 2006) tried to tackle the problem using tracking techniques, but the idea was rapidly dropped because of the amount of the clutters and occlusions in the scenes, which makes it impossible to keep track of foreground objects. On the other side, background subtraction based methods achieved a good recall, nevertheless they may present false alarms when it comes to a scene with many sources of noise (crowds, sudden illumination changes...).

The robustness of such a system depends on:

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- The effectiveness of the background subtraction method: A minimum of noisy outputs (illumination changes, ghosts, noise), can greatly reduce moving object related problems.
- The classification method: Using robust adaptive scores, and cues may reduce the false alarms rate considerably.

In this paper, we propose a novel method that uses edges to detect and classify AO candidates. Our focus is on reducing false alarms in complex scenes. The proposed method is robust to illumination changes, and can accurately reject false alarms resulting from still or sitting persons. We detect stable edges by applying temporal accumulation on the output of a foreground edge detection method. These stable edges are then grouped using a clustering algorithm to form bounding boxes of AO. Next, in order to classify the AO candidates, we propose a robust score based on the configuration and orientation of the AO candidates' edges to check whether the bounding box really encloses an object or not. Another score based on static edges consistency is proposed to check the object stability in time, and to reject candidates with small and internal movements like still persons.

The main contributions of this paper are as follows: Edges are used instead of pixel intensities for static regions detection. This choice is motivated by the fact that edges are more robust to illumination changes which are a major source of false alarms. Our contribution consists in using edges in the classification step, and not only in the detection step, as opposed to Kim et al. (2014). Furthermore, an accurate clustering algorithm is used to group static edges using their position, and the time of being static, unlike Kim et al. (2014) who used only the position to group stable edges. Finally we propose two robust scores for classifying the static region candidates into true or false positives. The first one is an objectness score, based on the edges position and orientation within the bounding box. The second one, is a staticness score, based on the resulting stable edges configuration.

For the rest of this paper, in Section 2, we present the state of the art of the static regions detection methods based on the background subtraction. Works which focus on classifying the AO candidates are also reviewed. Tracking based methods are reviewed briefly. In Section 3, the proposed method is presented in detail with an explanation of the scores. Section 4 summarizes the experiments and the obtained results.

2. Related works

The tracking-based methods (Auvinet et al., 2006; Guler et al., 2007; Krahnstoever et al., 2006; Martínez-del Rincón et al., 2006), have been used to detect abandoned objects by analyzing the motion history of the foreground objects, using single or multiple view cameras systems. As stated before, the idea was dropped because of the complexity of the scenes. Therefore, most of the efforts were concentrated on approaches using background subtraction to detect static regions. They rely almost all on the GMM (Gaussian Mixture Model) proposed first by Stauffer and Grimson (1999), where a pixel intensity change is tracked in time, by a set of k Gaussians. If the input pixel is checked to belong to one of the Gaussians then it is classified as a background pixel, otherwise it is classified as a foreground pixel and a new Gaussian is created for that pixel using its value as its mean. Every approach has its own strategy, and can be divided into three categories. The first category uses Dual background modeling. In Porikli et al. (2007), the authors used two background subtraction models with different learning rates, and two foreground masks F_{short} (short-term foreground) and F_{long}(long-term foreground) are generated. Abandoned objects are simply identified by regions where F_{short} pixels are 0 and F_{long} pixels are 1. Wahyono et al. (2016) also, used two background models complemented with an object detector module to identify true AO. In Evangelio and Sikora (2010), the authors tried to improve the concept by integrating a finite state machine in order to monitor the pixel evolution in time and to classify the pixel according to the final state. Lin et al. (2015) continued with the idea, and in addition they used a back-tracing tracker of the owner to say whether the candidate is an abandoned object or not. Moreover, Wang et al. (2014) combined a background subtraction method with a technique for motion computation, to generate two moving objects masks, where pixels with value 1 in the foreground mask and value 0 in the motion computation mask are classified as static. The second category uses the GMM properties. Tian et al. (2011) used the GMM properties to detect relatively static pixels. they used a GMM of three Gaussians , and a pixel is classified as static by simply detecting its transition from a low weighted Gaussian to a high weighted Gaussian. The third category involves the use of the foreground masks sampling. In Liao et al. (2008) the authors used a foreground sampling accumulation to identify static regions, foreground masks are accumulated every 100 frames for 30 s using an *AND* operator.

Detecting static regions by classifying pixels individually, can generate a noisy static mask. Thus, working at pixel level is not sufficient, and requires some further processing on those formed static regions. Pan et al. (2011) proposed a region level analysis to reduce the false alarms by defining two scores; the foregroundness, intended for background maintenance by checking every foreground blob, and see whether it is caused by illumination changes or ghosts, and the abandoness used to identify the static regions. Szwoch (2016) proposed a technique that checks foreground pixels stability in time. Then, a clustered region of these stable pixels is compared with contours of moving objects resulting from the background subtraction step to reduce false alarms from resulting from spurious detections.

Moreover, Hassan et al. (2013) focused on dealing with illumination changes, especially for outdoor stationary object detection by defining a contour tracking score for the AO candidates. An attempt was done in Kim et al. (2014) to detect static abandoned using an edge-based background subtraction model. The method was tested on the publicly available dataset PETS2006 only and presented some false alarms in the second scenario.

Some other works focused more on reducing false alarms, by adding a classification module as a post-processing step. The shape of the object, the context, and time were used as features to classify AO into true or false positives. In Fan et al. (2013) three attributes where used: The foregroundness, the staticness, and the abandonment which define respectively the likelihood that an object be a foreground object, the likelihood that an object be static for a defined period of time, and the likelihood that an object be left by its owner. These attributes were used as high level features to detect true abandoned objects. Each one of these attributes was trained using low-level features. Using these quantified attributes, noise was reduced, and adaptability was increased. Similarly Kim and Kim (2014), used three features (intensity, motion and shape) to identify abandoned objects, achieving a good detection rate. They also proposed in Ortego and SanMiguel (2014) a spatio-temporal set of features to check the stationarity of a region.

3. Proposed method

We propose a novel edge-based approach for AO detection and classification. Fig. 1 shows the proposed system architecture. There are three main components:

a) Stable edges detection.
b) Stable edges clustering and orientations extraction.
c) AO candidates classification.

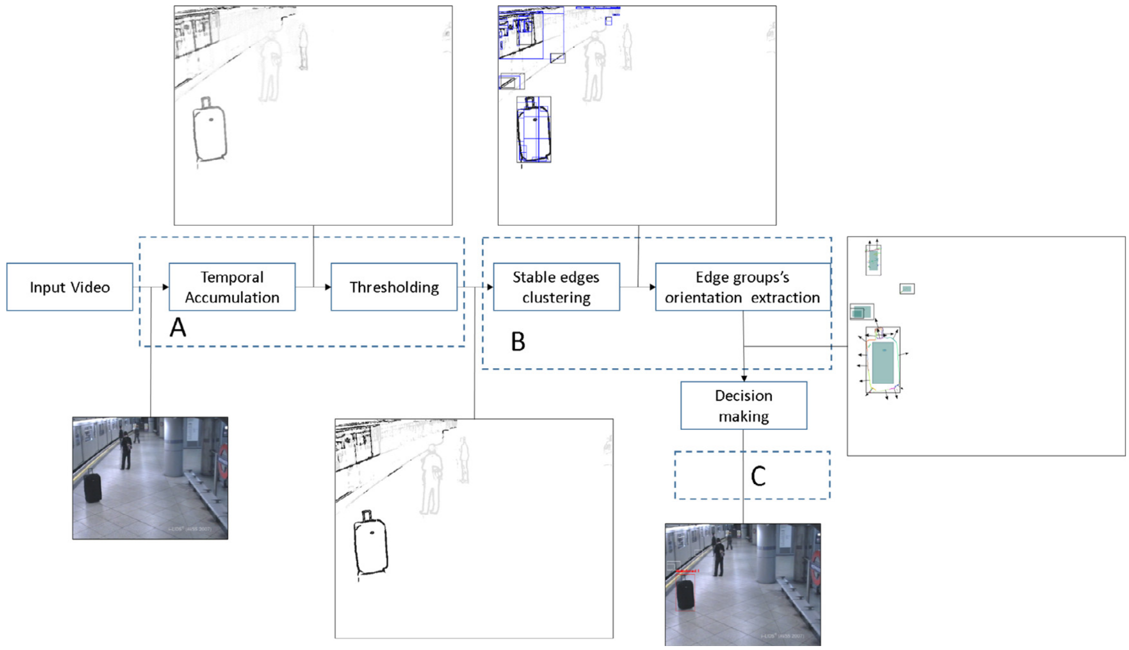


Fig. 1. Flowchart of the proposed method. A:stable edges detection. B: stable edges clustering and orientations extraction. C: AO candidates classification.

For stable edges detection, a moving edge detection technique, is combined with a temporal accumulation process to generate a stable edges mask. Using edges instead of pixel intensities was motivated by the fact that edges are more robust to illumination changes, require no shadow removal, and describe the scene in a better way. The resulting edges from the stable edges mask are clustered using position and time to delineate the supposed abandoned object bounding box. Then, the edge orientations within a bounding box are calculated. Finally we extract an objectness score using the edges orientations to check if the bounding box really encloses an object. The main motivation, is that by only checking whether there is an object inside a bounding box or not, we can filter almost all false alarms like ghosts which result in a randomly and fragmented stable edges configuration.

Moreover a staticness score is extracted to eliminate candidates with small and internal movements, like a still person. It is calculated by taking advantage of the resulting stable edges configuration within a bounding box. In fact temporal accumulation is more sensitive to edges than pixel intensities i.e. any small or partial movement will generate a fragmented, non-consistent edges configuration on the stable edges mask. Thus, the score is computed by simply measuring these defects in stable edges configurations. Finally we define thresholds for the two defined scores to make a decision on an AO candidate, as a true or a false positive AO. The orientations are extracted immediately after the bounding box generation, and require no more further observation.

3.1. Stable edge pixels detection

Various methods for background subtraction based on edges have been proposed in the literature (Davis and Sharma, 2007; Gruenwedel et al., 2011; Kim et al., 2013). In this work we employ the technique proposed by Gruenwedel et al. (2011) to separate moving edges from the background image. The method operates on the X and Y directions independently. i.e. the background is estimated in X and Y independently. First, the Sobel operator is applied on the input image to compute The first derivatives in both X and Y directions. Then, using the running average, the gradient of the background scene for each direction is estimated, over time

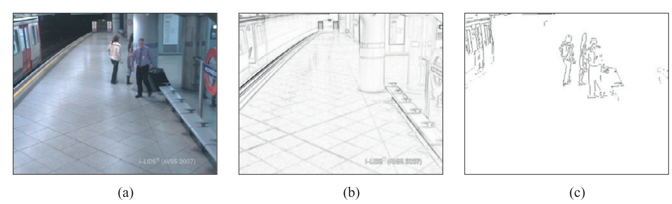


Fig. 2. Moving edges detection a). Input frame b). Background image c). Foreground image.

and at each pixel(Eq. (1)). The update of the background model is performed using the difference between the input gradient and the background image gradient(Eq. (2)). Finally, the gradient of the input image in each direction, is compared with the background model to generate a binary foreground mask for the respective direction(Eq. (3)).

$$B_{x,t}(x,y) = B_{x,t-1}(x,y) + \alpha D_{x,t}(x,y) \tag{1}$$

$$D_{x,t}(x,y) = G_{x,t} - B_{x,t}(x,y) \tag{2}$$

$$F_{x,t}(x,y) = hyst(|D_{x,t}|, T_{low}, T_{high}) \tag{3}$$

where $B_{x,\,t}$, $D_{x,\,t}$, $F_{x,\,t}$, $G_{x,\,t}$ and α are respectively the background model in the X direction, the difference between the background model and the current gradient image, the binary foreground edges mask, the gradient difference in the X direction, and the learning rate(the same applies for the Y direction) (Fig. 2).

$F_{x,\,t}$ and $F_{y,\,t}$ are calculated using a hysteresis thresholding function. Pixel values higher than T_{high} are set to 1, while those lower than T_{low} are set to 0. However pixel values between T_{low} and T_{high} are set to 1, if one of the pixels in the 8 connected neighbors has a value greater than T_{high}, otherwise they are set to 0. Finally the foreground edges mask F is obtained using an *OR* operator of $F_{x,\,t}$ and $F_{y,\,t}$. Unlike the authors in Gruenwedel et al. (2011) who use a short-term model with a high learning rate combined with a long-term model with a low learning rate to suppress noisy and unwanted moving edges, in this work, and for simplicity, we use only one model to detect the moving edges.

Fig. 3. Static edges accumulation in time at frames 1110, 1570 and 2070.

We detect stable edge pixels using the binary foreground mask F_t and a temporal accumulator for each edge pixel:

$$ACC_t(x,y) = \begin{cases} ACC_{t-1}(x,y)+1 & \text{if } (F_t(x,y)=1 \ \& \ i\%10=0) \\ ACC_{t-1}(x,y)-1 & \text{if } (F_t(x,y)=0 \ \& \ i\%10=0) \end{cases} \tag{4}$$

$$SEMask_t(x,y) = hyst(ACC_t(x,y), AO_{time}/2, AO_{time}) \tag{5}$$

ACC is the temporal accumulation mask of stable edge pixels initially set to 0, and is accumulated only where edge pixels are detected ($F_t(x,y)=1$). Moreover, the condition ($i\%10=0$) in Eq. (4) was incorporated to avoid unwanted moving edges accumulation of pixels, in other words, to avoid taking into account temporally static and slow motion objects. So instead of doing frame by frame accumulation, *ACC* is updated every 10 frames. Finally we obtain stable edges mask *SEMask* by applying a hysteresis thresholding function on *ACC*(Eq. (5)). Using a hysteresis thresholding instead of single value thresholding helps overcome partial occlusions and slow moving objects, that prevent static edge pixels from accumulation in the *ACC* mask. AO_{time} is the duration threshold for abandonment time.

Stable gradient estimates for both directions *X* and *Y*are obtained using the binary mask *SEMask*(Eq. (6)). Stable thin edges are extracted by simply applying non-maximum suppression(NMS) on $Sg_{x,\,t}$, and $Sg_{y,\,t}$.

$$Sg_{x,t}(x,y) = \begin{cases} G_{x,t}(x,y) & SEMask(x,y)=1 \\ 0, & otherwise \end{cases}$$
$$Sg_{y,t}(x,y) = \begin{cases} G_{y,t}(x,y) & SEMask(x,y)=1 \\ 0, & otherwise \end{cases} \tag{6}$$

3.2. Edge clustering

To generate the AO candidates bounding boxes(BB), edges resulting from *SEMask* are grouped using their spatial and time information. Each edge segment is enclosed in a rectangle, and its stability is tracked in time to be used as temporal distance in clustering. For spatial distance between rectangles, we use the minimum distance between the four corners of a rectangle with those of another, instead of using the rectangles centroids. A recursive labeling technique, is proposed for the clustering. The algorithm traverses the non-labeled rectangles recursively as a graph in order to transfer the edge labels. As mentioned above, using the time of stability and the spatial distance, each rectangle propagates its label to the non-labeled rectangle in its neighbor if the distance and time thresholds D_{th} and T_{th} are satisfied. The pseudo-code is listed in Algorithm 1.

Fig. 3 shows an example of the edge segments grouping; the bottom row shows the *SEMask* map evolution in time at frames 1110, 1570, 2070. Static edges are enclosed in the blue boxes, and the black boxes are the abandoned objects candidates, resulting from clustering edge boxes with the same label. When using the T_{th} threshold for staticness time, the algorithm works well in cases of close abandoned objects or even overlapping objects with different times of drop, by generating two bounding boxes for each object.

Algorithm 1 Pseudo code listing of the clustering algorithm.

```
counter ← 0
Initialize all n Rectangles labels to 0
procedure GROUPING
    for k ← 0, n do
        PROPAGATE(k)
    end for
end procedure
procedure PROPAGATE(i)
    for j ← 0, n do
        if (j ≠ i && Rect_j.label = 0) then
            if (dist(i, j) < D_th && Rect_i.time − Rect_j.time < T_th) then
                if Rect_i.label = 0 then
                    Rect_i.label ← counter
                    Rect_j.label ← counter
                    Counter++;
                else if Rect_i.label > 0 then
                    Rect_j.label ← Rect_i.label
                end if
                PROPAGATE(j)
            end if
        end if
    end for
end procedure
```

3.3. Classification

Already proposed methods (Kim and Kim, 2014; Pan et al., 2011), usually employ numerous scores to check if the AO candidate is either caused by a ghost, sudden illumination changes, still persons. A score is defined for each of these false detection cases. However in our work, instead of checking each false detection, we classify an AO candidate into a true or false abandoned object by checking only if there is an object enclosed within the bounding box. We were motivated by the recent success of category-independent object detection methods (Cheng et al., 2014; Endres and Hoiem, 2010; Rahtu et al., 2011; Zitnick and Dollár, 2014), where the goal is to spare for object detection techniques to check all possible positions and scales of the sliding window by generating a set of regions, or bounding boxes proposals with high probability of being an object, prior to applying object detection. In this work, we use this concept as a post-processing instead, and the aim is to check whether there is an object enclosed within the bounding box or not, regardless of its category.

3.3.1. Objectness score

For the human eye perception system, things with well-defined closed boundaries are likely to be objects (Cheng et al., 2014; Endres and Hoiem, 2010; Rahtu et al., 2011; Zitnick and Dollár, 2014). We take as assumption that an abandoned object or luggage has in general simple convex boundaries, in defining our objectness score. Zitnick and Dollár (2014) propose a method for object proposals that score a bounding box by the quantity of edges enclosed within. They developed a technique that weights edges according to their affinities to the edges that directly overlap bounding box

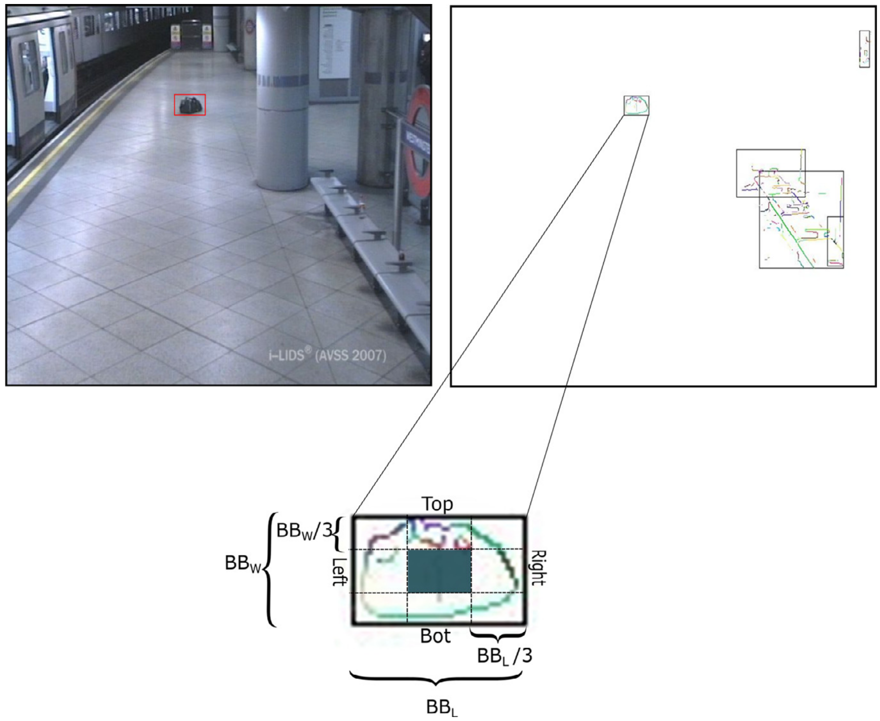


Fig. 4. Dividing of bounding box in regions.

boundaries. Their objectness scoring of the bounding box is by using these edges's weights. An efficient data structure was used for representing edges by grouping edge pixels of the same orientation into groups, then affinities are computed between these edge groups. Finally they are weighted according to their affinities to edges groups which directly overlap the bounding box boundaries. The objectness score proposed in Zitnick and Dollár (2014) works well for simple high scale images with clear background, but it is not suited for complex surveillance videos, because of the complexity of the scenes and sizes of AO candidates that can be very small.

We propose a suited objectness score, that is more adequate for the luggage objects case. A bounding box of an AO candidate BB with width BB_W and length BB_L, is divided into four regions Left, Right, Bot, and Top, as shown in Fig. 4. Next, using the edges orientations, the objectness score is computed by measuring the enclosed object convexity in each of these regions. To do this, we use the edge group representations and their average gradient directions as in Zitnick and Dollár (2014), where edge groups are formed by grouping connected edge pixels according to their orientations until the sum of their gradient orientations difference exceeds $\pi/2$.

$$T_{reg} = T_{reg} + segLength_i \quad \text{if } |\sin(\theta_i)^2 - 1| < \sigma \tag{7}$$

$$B_{reg} = B_{reg} + segLength_i \quad \text{if } |\sin(\theta_i)^2 - 1| < \sigma \tag{8}$$

$$R_{reg} = R_{reg} + segLength_i \quad \text{if } |\sin(\theta_i)^2| < \sigma \tag{9}$$

$$L_{reg} = L_{reg} + segLength_i \quad \text{if } |\sin(\theta_i)^2| < \sigma \tag{10}$$

where θ_i and $segLength_i$ are respectively the mean orientation and the length of the i^{th} edge group in each region.

T_{reg}, B_{reg}, R_{reg} and L_{reg} are respectively the sums of the edge groups lengths that satisfy the convexity condition in Top, Bot, Right and Left regions (Fig. 5). For Top and Bot regions, lengths

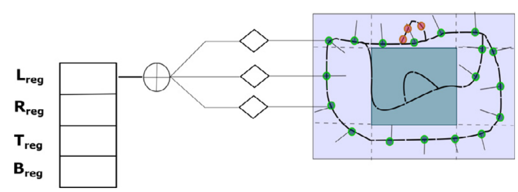


Fig. 5. Illustration of the gradient direction of the AO candidate edges groups. Edge groups with green circle on the left region satisfy Eq. (10).

of edge groups with a mean orientation near β=($\pi/2$ or $3\pi/2$) are accumulated when ($|\sin(\theta_i)^2 - \sin(\beta)^2| < \sigma$) (Eq. (7) and (8)). For Right and Left regions, lengths of edge groups with a mean orientation near β=(0 or π) are accumulated when ($|\sin(\theta_i)^2 - \sin(\beta)^2| < \sigma$). (Eq. (9) and (10)).

Finally we compare T_{reg} and B_{reg} with the bounding box length BB_L, and compare R_{reg} and L_{reg} with the bounding box height BB_W, by defining the following score S_b:

$$S_b = \frac{\lambda}{((L_{reg} - BB_W) * (R_{reg} - BB_W) * (T_{reg} - BB_L) * (B_{reg} - BB_L))^2} \tag{11}$$

λ is being a constant.

3.3.2. Staticness score

False alarms can be generated from still persons or objects with small and internal movements. To cope with that, we propose a score to check the object's boundary stability in time. We calculate a staticness score by measuring the contours stability of the object. Edges have high sensitivity to movement and therefore it is easy to capture any small and discrete movement. Formally, Bounding boxes in *SEMask*, containing highly fragmented stable edges are unlikely to be abandoned objects. We calculate a score C_b by measuring the object contour connectivity by verifying the connections of each edge group with its neighboring ones (Eq. (12)). We use the assumption that for an object with a simple boundary, each

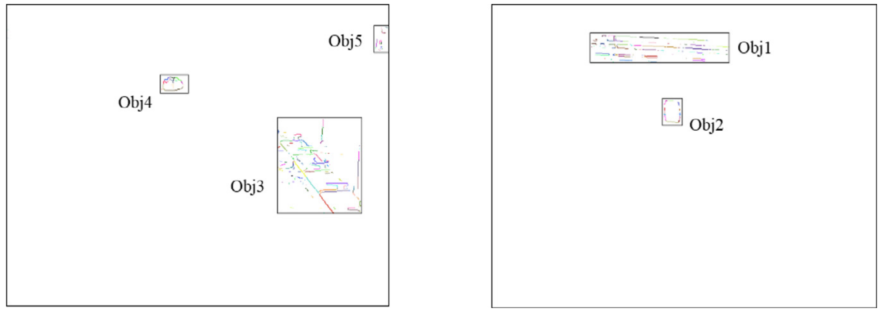


Fig. 6. Scores of different AO candidates from PETS2006 S2 scenario (right) and AVSS2007 medium video(left) :Obj1 (S_b=0,126.10-12, C_b=0) Obj2 (S_b =0,0008, C_b =0,00018) Obj3 (S_b =2,123.10-5, C_b =0) Obj4 (S_b =8,92, C_b =3,796) Obj5 (S_b =1,656.10-21, C_b =1,5652.10-5)

edge group has at least two connections with its neighboring edge groups, and so edge groups with less than two connections will penalize the score. The proposed score helps filter false alarms resulting from non-mobile objects, but with internal movements like still persons.

$$C_b = \prod \frac{|\phi(i)|}{2} \tag{12}$$

$\phi(i)$ is the set of inter-edges connections for the i_{th} edge group.

Fig. 6 shows some objectness and staticness scoring examples of some AO candidates. *Obj*1 is a ghost left by the garbage removed from the scene and therefore highly fragmented random edges with no defined boundaries are clustered in a bounding box, the objectness and staticness are too low and is rejected from being an abandoned object. On the other hand, *Obj*4 and *Obj*2, which are both objects and remained static for a defined period of time, have high S_b and C_b scores. Thresholds for both scores will be defined in the next section. The C_b score is effective in cases of static humans seen as AO candidates, with the fact that even in a static state, a human makes some small internal movements. This results in highly fragmented edges because edges are sensitive to small movements.

We use two thresholds *T*1 and *T*2 for the two scores S_b and C_b respectively. If S_b is greater than T1 and C_b is greater than T2, then the inside object is validated and considered as a true abandoned object. To avoid taking into account moving objects edges occluding the AO candidate, only the edge distributions with an accumulation value higher than $(AO_{time}/2)$ are considered in scoring.

4. Experiments & results

4.1. Experimental setup

The algorithm was implemented in C++ using OpenCV for preprocessing and capture routines, on a general purpose laptop, with an I7 CPU. The method is evaluated on the publicly available datasets used in the abandoned object detection literature (Cuevas et al., 2016): I-LIDS's AVSS2007 , PETS2006, and PETS2007. Moreover the method was also tested on some newly introduced state of the art benchmarks as CDNET2014, and ABODA datasets. The evaluation is performed using Recall, Precision and F-measure. Recall is a ratio of true detection to the ground truth, while Precision is a ratio between the number of true detections and the number of all detections. F-measure is the harmonic mean between Recall and Precision.

$$\text{F-Measure} = \frac{2 \times Recall \times Precision}{Precision + Recall}$$

Table 1 summarizes the parameter values used in the algorithm. $T_{low} = 40,\ T_{high} = 70,$ and $\alpha = 0.005$ are for the background maintenance. They were chosen in a way that the absorption time of

Table 1
Summary of the parameters used in the algorithm.

Parameter	α	T_{low}	T_{high}	σ	T1	T2
value	0.005	40	70	0.5	10^{-5}	10^{-5}

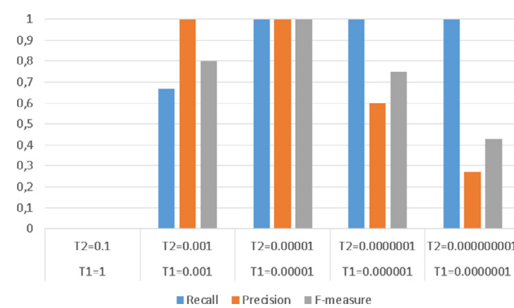


Fig. 7. Algorithm evaluation on AVSS2007 using different threshold values.

an object in the background model is greater than the abandonment time duration AO_{time}. AO_{time} is a user-defined parameter. The σ threshold for an edge group to be taken into account in scoring was set to 0.5, which means that the difference in orientations between an edge group, and its relative bounding box's bound should be inferior than $\pi/4$. λ is a constant and is set to 10^8. Thresholds *T*1 and *T*2, for S_b and C_b respectively, were decided by conducting experiments on AVSS2007 videos, to find the optimal cutoff between the true positives class and the false positives class; see Fig. 7. Five threshold configurations were chosen to decide the best cutoffs for both scores. Performance of each configuration is measured using Recall, Precision, and F-measure.

Fig. 7 shows that when the values decrease for T1 and T2, Recall is high, but precision decreases, which means that all positives are classified correctly, but there is a high false positive rate, so decreasing T1 and T2 means more tolerability. On the other hand, for higher values of T1 and T2, Recall decreases because not all true positives are detected, while precision is high and consequently the classification is conservative. We observe that the configuration T1 = 10^{-5} and T2 = 10^{-5} results in the best performance, with precision and recall both equal to 1.0. Consequently, these values were chosen as thresholds for the algorithm. It is to note that the thresholds for classification are the same for all datasets, and needed no further tuning.

4.2. Results & discussion

4.2.1. Benchmark's results

PETS2006. The dataset contains seven videos from a train station, with a four angle multi-camera system to monitor the place. We have chosen the third camera angle for our algorithm, the videos

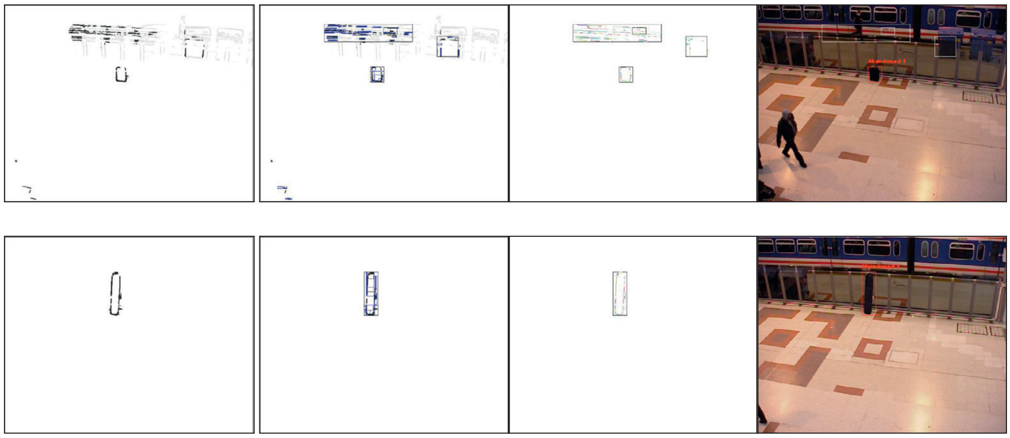


Fig. 8. PETS2006 abandoned object detection examples.

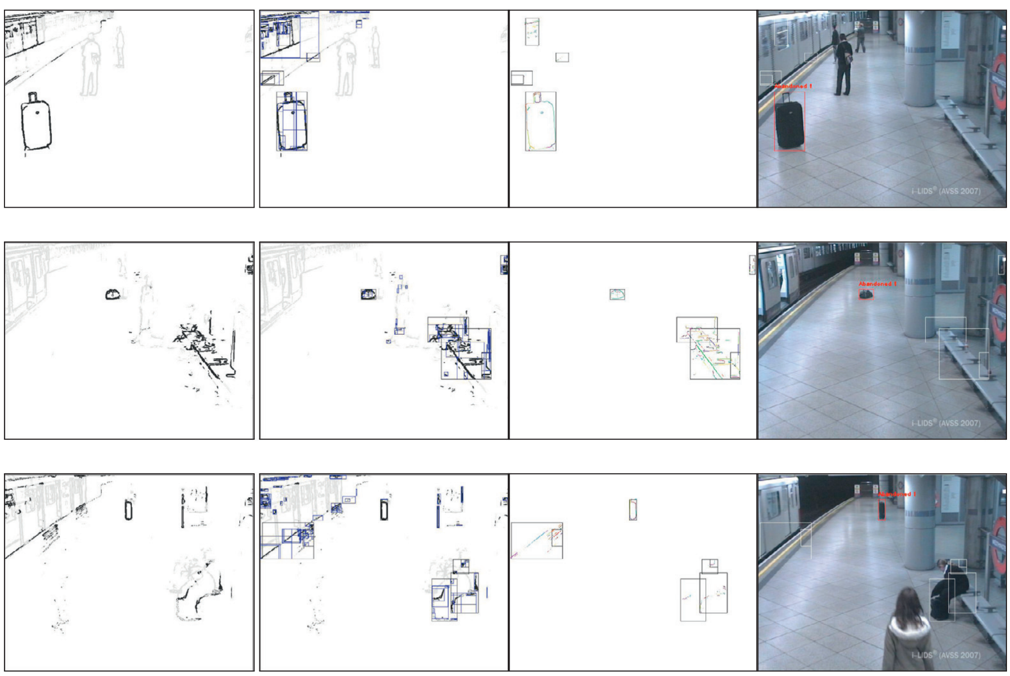


Fig. 9. AVSS2007 abandoned object detection examples.

contain simple events of people putting objects on the ground and leaving them behind. In the S2 scenario video, our algorithm performs better than those presented in Kim et al. (2014), where a region resulting from a removed object (garbage) is misclassified as an AO, and the bounding boxes were not accurately delineated for objects. Examples of AO detection are shown in Fig. 8. The first column shows *SEMask*, while the second column shows the *SEMask* mask after clustering. The third column shows the edge groups representation for scoring, and the forth column shows the AO object.

AVSS2007. The dataset contains three single view videos of a metro terminal with different difficulty levels: Easy, Medium and Hard. The difficulty is defined by the distance of the region where the object is left, its size, and the quantity of crowds occluding the AO candidate. Examples of detections are illustrated in Fig. 9.

PETS2007. The dataset contains videos with loitering, abandoned and re-attended baggage events. Both The S7 and S8 scenarios contain abandoned object events. These videos are a real challenge to the algorithm in terms of local and global sudden illumination changes, in addition to the quantity of crowds. Examples of our AO detection are shown in Fig. 10. Bounding boxes enclosing random edges distribution are results of strong illumination changes that affect the edges thickness, and thus leading to an accumulation in *SEMask* resulting in AO candidate boxes. However, the effectiveness of our classification module prevents these false alarms by classifying them as non-object candidates.

4.2.2. Quantitative results

A quantitative comparison on PETS2006 and AVSS2007 datasets, was performed with recent state of the art techniques (Fan et al., 2013; Lin et al., 2015; Pan et al., 2011; Szwoch, 2016; Tian et al., 2011). From Table 2, we observe that almost all methods detect abandoned objects accurately, except methods in Fan et al. (2013), and Szwoch (2016), which present missing detections in PETS2006, with a Recall equal to 0.8 for Fan et al. (2013) , and 0.86 for Szwoch (2016). On the other hand, the method proposed in Tian et al. (2011) detected all objects. Yet, there were false detections in the AVSS2007 scenarios, consequently their Precision and F-measure are lower than those obtained in the other methods. Both

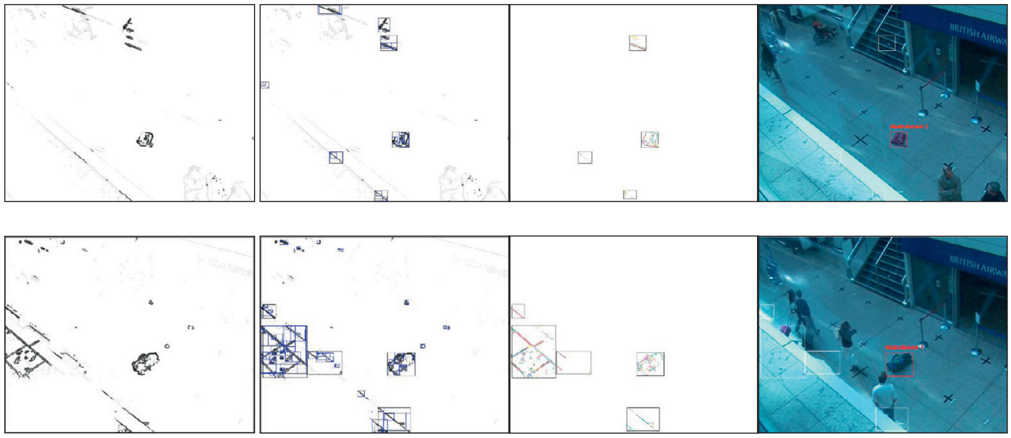


Fig. 10. PETS2007 abandoned object detection examples.

Table 2
Comparison results on PETS2006 and AVSS2007.

Method	PETS2006			AVSS2007		
	R	P	FM	R	P	FM
Tian et al. (2011)	1.0	0.85	0.92	1.0	0.35	0.52
Pan et al. (2011)	–	–	–	1.0	1.0	1.0
Szwoch (2016)	0.86	1.0	0.98	1.0	1.0	1.0
Fan et al. (2013)	0.8	0.95	0.87	1.0	0.97	0.98
Lin et al. (2015)	1.0	1.0	1.0	1.0	1.0	1.0
P.Approach	1.0	1.0	1.0	1.0	1.0	1.0

Table 3
Processing time complexity of the proposed algorithm.

Module	720 × 576	320 × 240
Stable edges detection	20 ms	4 ms
Clustering	2.4 ms	0.7 ms
Orientations extraction	32 ms	4.5 ms
Overall System	54.4 ms	9.2 ms

Lin et al. (2015), and our method performed perfectly in both datasets, with F-measure=1.0. However, in Lin et al. (2015) the authors have restricted their detection area to the train station platform in AVSS2007, and to the waiting zone area in PETS2006 only. In contrast, our algorithm performs, all over the scene in both datasets. Thanks to our Objectness and Staticness scores, our algorithm is capable to filter non-static, luggage objects. The method presented by Pan et al. (2011) have rough results in AVSS2007 datasets, however it was not evaluated on PETS2006 or PETS2007.

To demonstrate the robustness of our method against GMM based methods, we performed an in-depth comparison with the method proposed in Tian et al. (2011). Fig. 11 shows a comparison at the static mask level, on the first camera of PETS2007 video S8, of our method with those proposed in Tian et al. (2011). The first camera of PETS2007 has the most difficult view angle of the four angles of the dataset, so we have chosen it to challenge the two methods in terms of strong illumination changes and quantity of crowds. The first row of Fig. 11 shows static masks of both methods at frame 705, while the second row shows static masks at frame 1780.

The main observation, is that our method produces a static mask with a minimum of noise and consequently, lower static region candidates are generated. Compared with our method, the algorithm proposed by Tian et al. (2011) yields a noisy mask because of sudden illumination changes, and the slow moving crowds. This results in high static region candidates as illustrated in Fig. 11. This can be explained by:

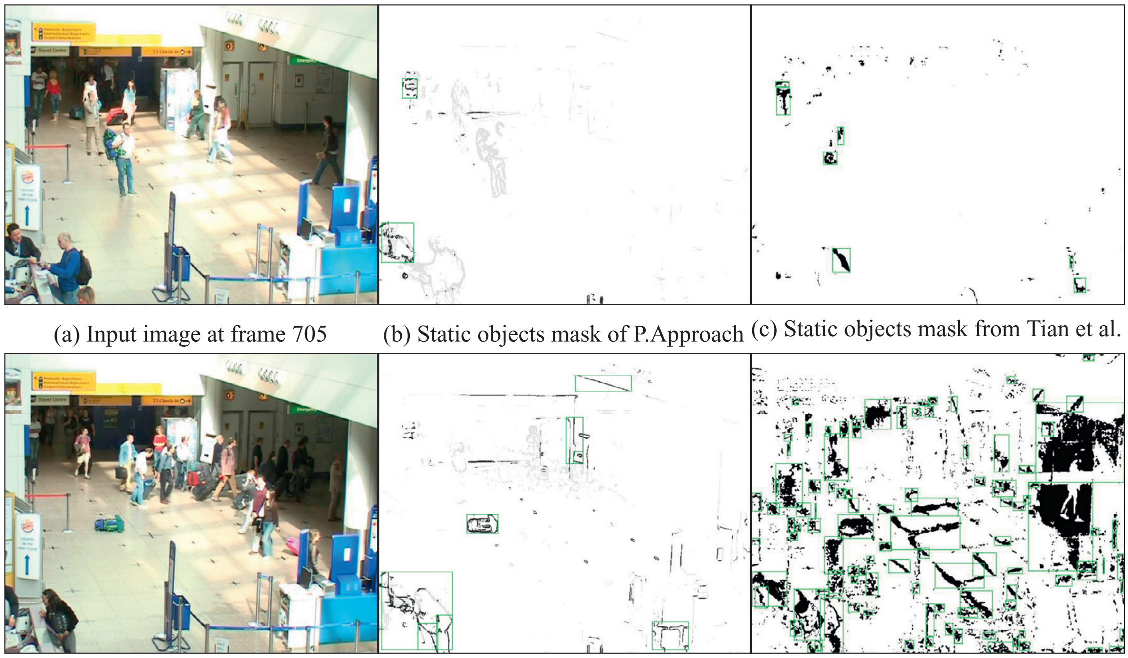
Firstly, in a cluttered environment(airport), with slow moving crowds, different objects with similar colors, will cross the same region, resulting in increasing the weight of the Gaussian representing foreground objects, pushing it into the Gaussians set, representing a background model; thus leading to a noisy static regions mask. Analogically, for our method, the same problem can cause false alarms.i.e. slow foreground moving objects crossing the same region can alter the temporal accumulation, resulting in noisy *SEMask*. However, this is avoided, since we use temporal accumulation at the edge pixels level. i.e. the probability that contours of two moving objects will overlap is very low.

Secondly, the GMM was originally designed to adapt to repetitive background motions(waving trees, cluttered background) (Stauffer and Grimson, 1999). Consequently, most of the time, the second Gaussian will represent periodic and random motions, rather than static objects, and this will lead to many small fragmented regions in the static mask. More specifically, in a GMM with three Gaussians, the two most weighted Gaussians(representing the background) are meant to model a periodic background motion composed of two objects, with different pixels intensities.

Another problem not addressed by Tian et al. (2011), is that different pixels of a recently static region can reach the second Gaussian at different speeds due to partial occlusion or similarities in the color of a region of the object with the background, and therefore small fragments of the static region will result in the static region mask, inducing more computations, or even false alarms. In our method this problem can also occur due to partial occlusion that prevents temporal accumulation in some parts of the static region, but this problem was solved using hysteresis thresholding to generate the static edges mask, as explained in Section 3.1. Finally, our solution is easy to implement, compared to Tian et al. (2011) which makes use of time-costly modules as: GMM, classifier-based human detector and tracking module.

4.2.3. Processing time

The processing time of the algorithm was analyzed using the AVSS2007and CDNET2014 datasets videos, which resolutions are 720 × 576 and 320 × 240 respectively. Table 3 shows the processing time of each module of the system. From Table 3 we observe that the edge group orientations extraction is the most time consuming module. On the other hand, clustering module is the fastest process and its processing time depends on the number of edges in *SEMask*. Finally, the stable edges detection module depends only on the image resolution. The overall computational speed is 18 Fps for 720 × 576, and 108 Fps for 320 × 240 resolution.



(a) Input image at frame 705 (b) Static objects mask of P.Approach (c) Static objects mask from Tian et al.

(d) Input image at frame 1780 (e) Static objects mask of P.Approach (f) Static objects mask from Tian et al.

Fig. 11. PETS2007's results comparison of our approach with Tian et al. (2011).

Table 4
Comparison of processing time with other methods, on 320 × 240 resolution.

Method	Processing speed
Ours	108 Fps
Szwoch (2016)	49 Fps
Lin et al. (2015)	29 Fps

Table 4 shows a processing speed comparison between our method and those in Szwoch (2016) and Lin et al. (2015) on 320 × 240 resolution datasets videos. Our approach clearly outperforms the compared methods. This can be explained by the following factors: Firstly, the used edge-based background subtraction method is much more simple, yet faster than the GMM used by most the state of the art methods including Szwoch (2016) and Lin et al. (2015). Secondly, since we use edges, no time consuming, or extra processing step is needed, as shadow removal. Moreover, our stable edges detection module generates *SEMask* with a minimum of noise, and thus less bounding boxes to process, which means less processing time.

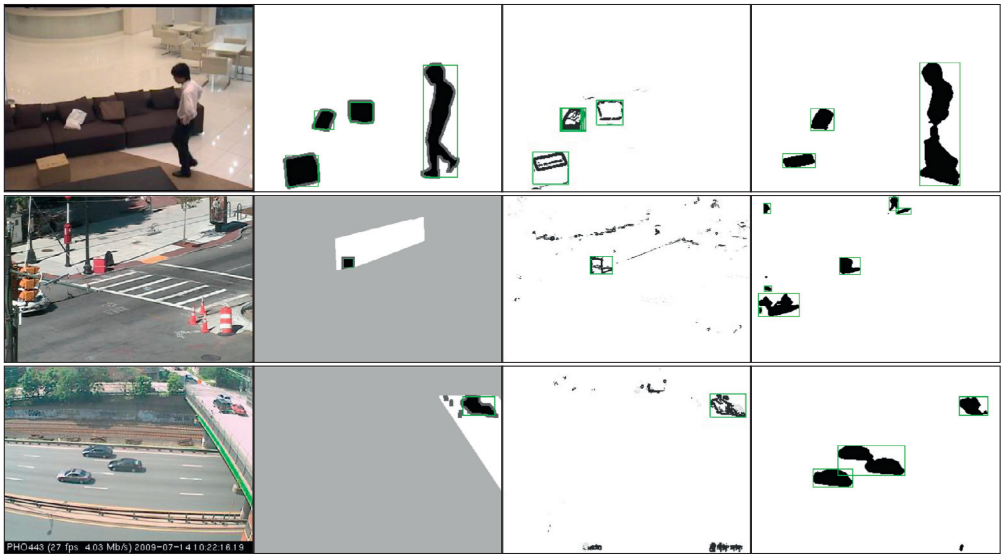
4.2.4. Results on newly introduced benchmarks

CDNET2014. The method was also tested on the CDNET 2014 dataset from the CVPR 2014 change detection challenge. The dataset is destined for the evaluation of motion detection methods; however it contains a category with intermittent object motion, with cases of abandoned and static objects. Since we use edges in detection, it was impossible to compare binary masks at pixel levels. Therefore, a connected component analysis was applied on the masks obtained in Wang et al. (2014) and comparison was done at the bounding box level. Results are shown in Fig. 12, hence only static regions detection step is used for comparison.

The first row represents a scene with 2 persons leaving 3 objects on a sofa. The method in Wang et al. (2014) has one missing detection, moreover the box object was not detected accurately because of the low gradient with carpet. On the other hand our method detects all abandoned objects accurately. The second row is an outdoor scenario with noise and hard illumination changes, where a box is moved from one position to another. Both the proposed and the one in Wang et al. (2014) detect the box accurately. The last row is also an outdoor scenario, where cars stop at streetlights. Both approaches detect the stopped cars. It is to note that The authors in Wang et al. (2014) proposed no module for discriminating still people from true AO, and thus their system can present false alarms when deployed in crowded places like transit zones.

ABODA. Furthermore we evaluated our method on the ABODA dataset proposed first in Lin et al. (2015). The dataset is constituted of 11 videos of abandoned luggage events, with different environment conditions; outdoor scenes, crowded scenes, and night-time detection. The dataset includes also scenes with light switching which results in a complete changing in the background image. Fig. 13 shows some detection examples from the ABODA dataset. The first column show *SEMask*, the second column shows the edge groups for the AO candidate and the last column shows the output image.

The first row shows an outdoor scenario where a person drops a bag and leaves the scene. The abandoned bag was successfully detected as abandoned object, without any false alarms. The second row shows a night-time detection where light is switched on. The algorithm was not affected by this sudden illumination changing, and all objects were detected accurately. The third row, shows a scenario of a classroom, where the light is turned off, then the camera switched to IR mode, affecting strongly the background scene. It results in a high number of AO candidates on *SEMask*. The abandoned object was detected with success but there were some false alarms. Table 5 shows comparison results, with Lin et al. (2015) and Wahyono et al. (2016). The method in Lin et al. (2015) and ours outperform approach proposed in Wahyono et al. (2016) in almost all videos of the dataset. On the other hand compared with Lin et al. (2015), our method did not present any false alarm in video 5, but it was not able to detect the AO in video 11 because of its small size.



(a) (b) (c) (d)

Fig. 12. Detection results on CDNET2014: first row: sofa scenario, second row: abandoned box scenario and the third scenario: streetlight scenario. a). Input frame b). Ground-Truth image c). Proposed method static mask d). Wang et al. (2014) detection mask image.

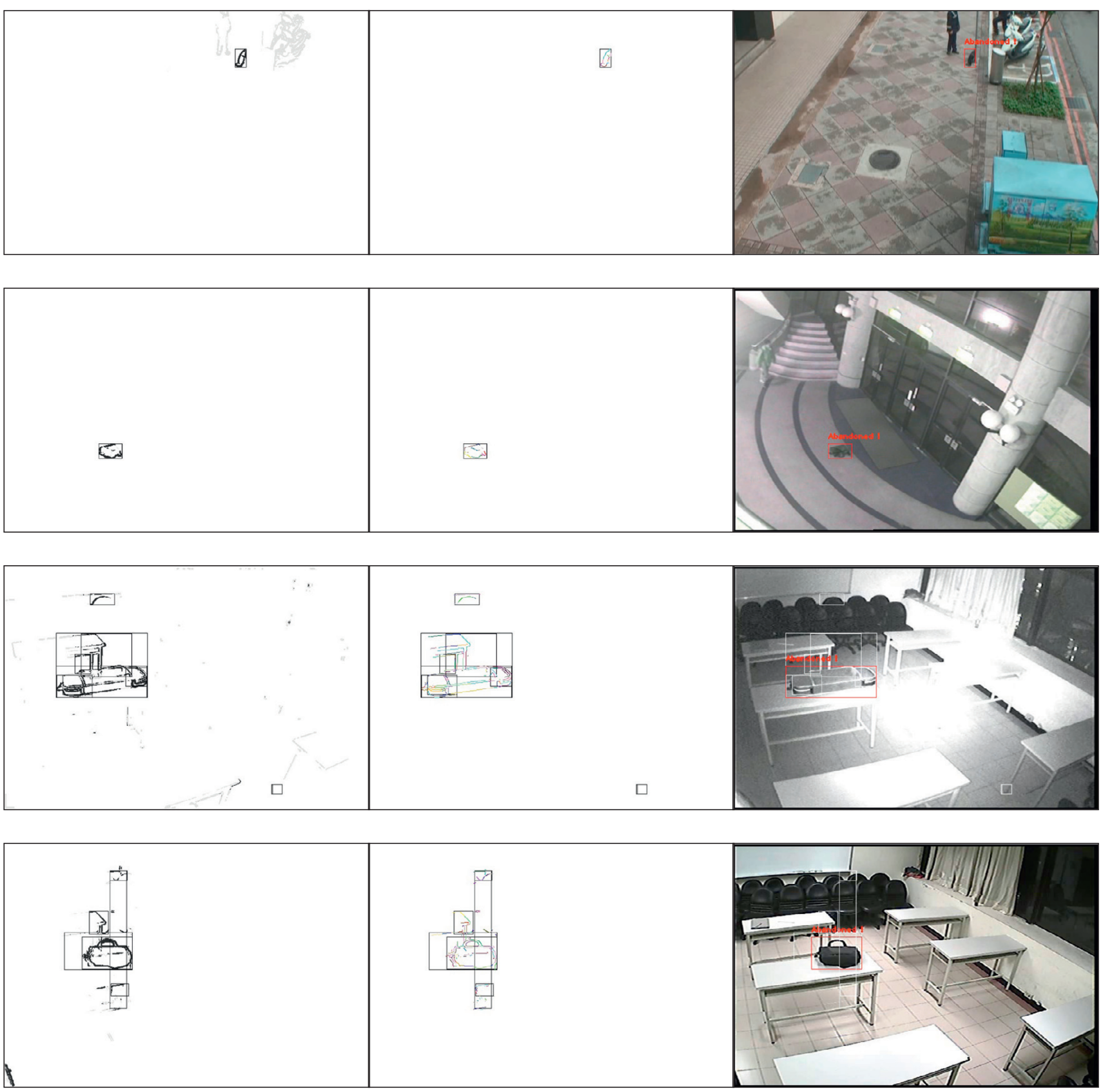


Fig. 13. ABODA abandoned object detection examples.

Table 5
Evaluation and comparison on the ABODA dataset.

Sequence	Scenario	Difficulty	GT	Lin et al. (2015)		Wahyono et al. (2016)		Ours	
				TP	FP	TP	FP	TP	FP
Video1	Outdoor	*	1	1	0	1	0	1	0
Video2	Outdoor	**	1	1	0	0	1	1	0
Video3	Outdoor	**	1	1	0	0	1	1	0
Video4	Outdoor	**	1	1	0	0	1	1	0
Video5	Night-time	***	1	1	1	1	0	1	0
Video6	Light Switching	*****	2	2	0	–	–	2	0
Video7	Light Switching	*****	1	1	1	–	–	1	2
Video8	Light Switching	*****	1	1	1	–	–	1	2
Video9	Indoor	*	1	1	0	1	0	1	0
Video10	Indoor	*	1	1	0	1	0	1	0
Video11	Crowded	****	1	1	3	–	–	0	1

5. Conclusion

In this work we have proposed a new method for abandoned objects detection based on edges in both the background subtraction and classification steps. We have used an edge-based background subtraction method for static edge detection by temporal accumulation. Edges are then grouped using time and connectivity information. A classification step is then performed to filter false detections by scoring the objectness of the AO candidates, and a staticness scoring for abandoned objects discrimination. The results obtained from the conducted experiments on different scenarios with different environment conditions, have shown that our method is indeed robust to sudden illumination changes, ghost effects, and other sources of noise. Some further work will be to incorporate an efficient detection and tracking module for the abandoned object owner.

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